Master Thesis — Progress Report

Rewiring the market clock in Spanish electricity markets

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Outline

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- Economic mechanisms: theoretical anchors and our mechanism
- What we want to test, data, and bid-shape evidence
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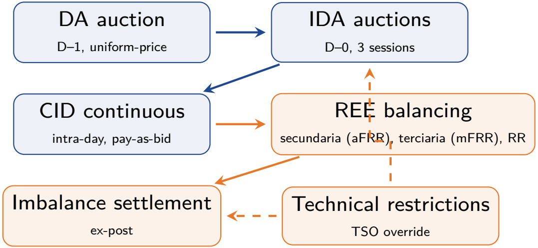
Motivation

- Empirically assess the impact of Spain's 2024–2025 **market time unit (MTU15) reform sequence** on *firm strategic behaviour* and *market outcomes*.
- Why interesting: the reform changes the time-unit granularity of the wholesale market in three stages, with *transitional asymmetric-clock windows* where settlement and trading run on different clocks. A useful setting to study how granularity shapes firm conduct.
- Why it is hard: the reform sequence overlaps with confounders — gas-price normalisation (Dec 2023), the 2025-04-28 blackout and post-blackout *operación reforzada* (reinforced operation), the elimination of CNMC Rule 28.8 for bilateral contracts (Mar 2025). **Institutional awareness is essential.**
- This presentation: identification strategy, headline result on **dominant firms** (Iberdrola, Endesa, Naturgy, EDP) strategic intraday repositioning, and an honest account of what survives robustness and what does not.

The Spanish wholesale market



- **OMIE (market operator)**: day-ahead, intraday auctions, continuous intraday.
- **REE (TSO)**: balancing markets (secundaria = aFRR, terciaria = mFRR, replacement reserve), imbalance settlement, technical restrictions.
- **Sequence:** forward markets get progressively closer to delivery; final adjustments via balancing.
- TSO can modify OMIE programmes when network or stability constraints bind.
- Firms have multiple instruments to take strategic positions across stages.

The 2024–2025 reform sequence

- Traditionally MTU = **60 min** everywhere.
- **Jun 2024 — IDA reform**: 6 local Iberian sessions consolidated into 3 European IDA sessions (still 60-min granularity).
- Three MTU15 reform stages:
  - **Dec 2024 — ISP15**: imbalance settlement → 15 min.
  - **Mar 2025 — ID15**: intraday → 15 min.
  - **Oct 2025 — DA15**: day-ahead → 15 min.
- Two transitional *asymmetric-clock* windows: settlement runs at finer granularity than trading.
- Plus: **2025-04-28 blackout** → *operación reforzada*, continuous from Apr 28 onward.

Period	ISP	ID	DA
Pre-Dec 2024	60	60	60
Dec24–Mar25	**15**	60	60
Mar25–Sep25	15	**15**	60
Oct25 onward	15	15	**15**

ISP15 — *Dec 2024*

ID15 — *Mar 2025*

DA15 — *Oct 2025*

Theoretical anchors and our mechanism

Allaz & Vila (1993, JET) — forward markets dispel rents

Adding more forward markets prior to the spot stage commits Cournot firms to quantities and erodes withholding incentives. *Caveat:* A–V adds **more markets for the same good** (e.g., a second forward stage). Our reform **divides one good into four sub-goods** (an hour into 4 quarters). **Related intuition; not the same mechanism.**

Ito & Reguant (2016, AER) — sequential markets and arbitrage

Firms with market power reposition across sequential markets (under-commit in DA, sell more in IDA). *Caveat:* I–R is about **sequencing across markets**; we are about **granularity within a market**. **Related, not equivalent.**

Chang (2026, WP) — settlement vs dispatch period mismatch

When settlement period differs from dispatch period, averaging effects change bidding incentives. *Does not apply to DA15/ID15*: settlement and dispatch are both 15-min — no averaging. Only the ISP15-with-DA60 transition window had averaging effects, and only on *quantities* (imbalance settlement averaged within the hour).

Our preliminary mechanism — within-market granularity (what we test)

Splitting an hour into 4 quarters that clear independently lets firms bid different prices and quantities across quarters of the same hour. **Especially valuable when residual demand varies steeply within the hour** (morning ramps, evening peaks). In flat hours where within-hour demand is constant, granularity has no economic content. **To be formalized!**

What we want to test, and how

Our mechanism predicts. Dominant firms exploit within-hour granularity in *critical* hours (steep within-hour ramps in residual demand) but not in *flat* hours (constant within-hour profile). The reform should differentially affect critical vs flat hours.

Outcomes:

- q_2 = dominant firms' *voluntary* IDA repositioning per firm-day-hour (signed PIBCIE), MWh. **Headline outcome.**
- q_1 = dominant firms' day-ahead auction-cleared sell, MWh per firm-hour.
- Bid-shape: tranche count, ladder slope, qty-weighted price (post-MTU15-DA only).

How we define critical and flat hours. For each hour-of-day h, compute $\sigma^2_{\text{within}}(h)$ = **the variance of conventional production (load − wind − solar) across the four 15-min quarters of hour** h, averaged over days. High σ^2_{within} means dispatchable units need to ramp a lot *within* the hour — exactly where within-hour granularity has economic content.

- **Treated** = critical hours $h \in \{7, 8, 16, 17, 18\}$. Top-5 by σ^2_{within} (morning ramp + evening peak).
- **Control** = flat hours $h \in \{3, 4, 5\}$. Bottom of σ^2_{within} (pre-dawn).

Why flat hours are a valid counterfactual. If the granularity reform has *zero economic content* in flat hours (because there is nothing within the hour to slice strategically), then comparing critical-hour behaviour to flat-hour behaviour identifies the mechanism. *This is a hypothesis I need to defend with a small theoretical model showing that, **in equilibrium**, the reform has no effect for flat hours — ongoing.*

Data

Coverage: 8 years (2018-01 → 2026-04), at native 60-min/15-min granularity throughout.

Three sources:

- **OMIE (Iberian market operator)**: prices and *bid-level* data for DA, IDA, continuous; per-firm and per-unit programmes. ∼5.7 billion rows (∼413 GB raw, ∼198 GB processed).
- **ENTSO-E Transparency Platform**: imbalance volumes and prices, balancing settlement, generation by type/unit, load and renewable forecasts. All system-level fundamentals come from here.
- **ESIOS (REE)**: Spanish per-segment settlement, real-time prices, balancing offers. **API token requested 3 weeks ago; still pending due to technical issues on the operator's side.** Reserves attribution at firm level partially blocked.

Engineering principles:

- Pipelines are **idempotent**: re-running any step produces the same output and includes new data — supports replication.
- 25+ data families, each with its own parser and tests.
- Empirical findings tracked formally (alive / wounded / dead) with a documented audit trail.

Honest caveat. Most of the empirical findings in the audit log are descriptive rather than carrying clear economic interest — e.g., the structural break in Q1 2024 nuclear bilateral share. The presentation focuses on the small subset that has both economic content and clean identification.

Bid-shape: dominant firms exploit granularity (justifies the design)

Direct evidence at the bid level that operators *actually use* within-hour granularity post-MTU15-DA — this justifies using flat hours as a valid counterfactual.

What we observe. For dominant-firm CCGT in critical hours (post-MTU15-DA):

- More price tranches per quarter (+8% vs flat hours).
- Steeper ladder slope (+13%).
- *Lower* qty-weighted average price.

Interpretation. Operators bid a richer ladder in critical hours: low-price inframarginal tranches at the bottom (push the qty-weighted average down, lock clearing) plus a steeper high-price tail (capture scarcity rent). They are unambiguously exploiting within-hour granularity.

Tech selectivity. Concentrated in CCGT and (mildly) hydro. Wind, solar, nuclear show no engagement. Crucially: many plants **mechanically repeat their previous hourly bid identically across all 4 quarters** — not exploiting the granularity at all.

Why this matters for the design. Bids confirm operators *can* and *do* use within-hour granularity in the right hours. Where granularity is exploited (critical), we should see effects on quantities; where not (flat), we should not.

The DiD specification

Panel. The unit of observation is (firm f, day d, hour h). We aggregate across the 4 quarters within an hour. So the outcome is q_{fdh}.

Specification (very preliminary):

$$q_{fdh} = \alpha + \beta_1 \, \mathsf{crit}_h + \beta_2 \, \mathsf{post}_d + \beta_3 \, (\mathsf{crit}_h \times \mathsf{post}_d) + \gamma_f + \delta_{\mathrm{DOW}(d)} + \varepsilon_{fdh}$$

β_3 is the DiD coefficient. γ_f are firm fixed effects (Iberdrola, Endesa, Naturgy, EDP); δ_{DOW} are day-of-week fixed effects. Cluster-robust SE by date d.

Window in each regression:

- **Pre** = pre-IDA: 2018-01-01 → 2024-06-13.
- **Post** = DA15/ID15: 2025-10-01 → present.
- Intermediate regimes (3-sess, ISP15-win, DA60/ID15) handled separately when needed.

Why the DiD removes time-varying confounders:

- Critical and flat hours come from the *same dates* ⇒ daily shocks (weather, demand level, fuel prices, regulatory announcements, macro conditions) absorb mechanically in the within-day comparison.
- Pre-IDA crit-flat differential serves as the parallel-trends sanity check.
- The contrast we identify is *mechanism-targeted*: hours where granularity should bite vs hours where it should not.

Heterogeneity. Most of the response carried by Naturgy (CCGT + Hydro). Iberdrola contributes through nuclear (caveat: allocation artefact). Endesa and EDP near-zero. Different technologies respond very differently — consistent with the bid-shape evidence on the previous slide.

Headline result: dominant firms reposition more in critical hours

Same panel and spec as previous slide: outcome q_{fdh} (signed IDA repositioning, MWh per firm-day-hour); $\text{crit}_h = 1$ for $h\{7, 8, 16, 17, 18\}$, 0 for $h\{3, 4, 5\}$; $\text{post}_d = 1$ from 2025-10, 0 for pre-IDA.

	Estimate	SE	p-value
β_1 (crit_h, pre-IDA crit−flat gap)	$+41.21$	3.36	1.6×10^{-34}
β_3 ($\text{crit}_h\times$ post_d, DiD)	$+58.67$	12.18	1.5×10^{-6}
$\beta_1 + \beta_3$ (DA15/ID15 crit−flat gap)	$+99.80$	—	—

$N = 78{,}177$ firm-day-hours; cluster-robust SE by date.

Type I / type II tradeoff. With $N \approx 80\text{k}$, statistical power is very high (low type II error), so the conventional $p < 0.05$ threshold becomes too lenient and inflates type I error. Worth thinking carefully about how to set significance thresholds in this regime.

Why these hours. The top-5 critical and bottom-3 flat are the hours where the robustness battery (per-hour DiD, cumulative critical-set sensitivity) confirms the effect most reliably — we use them as the cleanest contrast.

Interpretation. The outcome q_2 (signed IDA repositioning above DA commitment) is the empirical signature **Ito & Reguant (2016) use to detect strategic intra-day arbitrage**: positive q_2 means firms under-commit in DA and resell in IDA. We adopt their measure but ask a *different* question: **does this strategic IDA repositioning grow more in critical hours after the reform splits each hour into 4 sub-periods?** The DiD coefficient $\beta_3 = +59$ says yes. Mechanism: **the reform lets firms bid different prices/quantities for each quarter of an hour. This pays off where the within-hour production profile is steep — so firms exploit the ramps. Flat hours have no within-hour variation, so no effect.**

Robustness checks (in words)

- **Same-calendar-month restriction.** Restricting pre-IDA to the same months as the post window (Oct–May) leaves the result at 96% of the magnitude. Calendar mix not the driver.
- **Baseline-window sensitivity.** Across 5 narrower pre-IDA windows, the within-DA15/ID15 fact (+100) is constant; reform-attribution component ranges +27 to +133. **To think about:** pre-trend in the crit-flat gap is not stable.
- **Per-hour decomposition.** Treating each hour individually shows two clusters with opposite signs (ramps + peak positive; midday negative). Justifies the 5-hour critical set.
- **Conditional parallel trends.** Wind + solar + demand $\times$ hour controls absorb only $\sim$ 13% of the pre-trend. *Does NOT hold.* **To think about:** an unobserved time trend (CCGT retirement, regulatory anticipation) is at play.
- **Bid-shape and extensive-margin.** Unit-level confirmation: richer bid ladders in critical hours (CCGT only); 13 CCGT units inactive in the asymmetric-clock window, return under *operación reforzada*.

What is robust, what is not

Robust:

- The within-DA15/ID15 fact: +100 **MWh more IDA repositioning in critical vs flat hours per firm-hour.**
- Bid-shape rich-ladder pattern in CCGT post-MTU15-DA.
- Two-cluster mechanism: ramp + peak positive; midday negative.

Not robust:

- The reform-attributable component (DiD δ) is a range, not a point estimate. Pre-trend instability is real.
- Conditional parallel trends with renewable + demand controls does not hold.
- Heterogeneity: Naturgy carries most of the result; cross-firm response is uneven.

The theoretical-model gap. The design needs a stylised IO model showing that, in equilibrium, the within-market granularity reform is a no-op for flat hours — this is what justifies using them as control. The model is anchored in *our* mechanism (bidding differently across the 4 quarters of an hour, exploiting within-hour ramps), not directly in Allaz–Vila or Ito–Reguant. **Out of scope**: a fully estimated structural auction model.

What is left

- **Stylised theoretical model** that justifies flat hours as control — the foundation of the empirical design.
- **Better identification strategy.** The clean flat-hour design came together yesterday; there is more thinking to do.
- Think about the demand side more — the current mechanism and design are supply-side focused.
- Include the continuous market in the analysis.
- Additional robustness checks once the mechanism interpretation is locked.
- Per-segment settlement once the ESIOS API token clears, but not necessary a bottleneck for the main result.

Thank you!